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Project Report on

“BILLING SYSTEM”

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Under the Guidance of

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Work carried out at



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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING



CERTIFICATE

This is to certify that the OOPS with JAVA project titled “**Billing system**” has been carried out by Harshita M (1WA24CS127), Himani P (1WA24CS129), Keerthi P(1WA24CS146), Mohit(1WA24CS171) during the academic year 2025-2026.

Signature of the guide

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DECLARATION

We, Harshita M(1WA24CS127), Himani P(1WA24CS129), Keerthi P (1WA24CS146), Mohit S D(1WA24CS171) students of 3rd Semester, B.E, Department of Computer Science and Engineering, BMS College of Engineering, Bangalore, hereby declare that, this project work entitled "Billing System" has been carried out by us under the guidance of **Pradeep sadanand**, Assistant Professor, Department of CSE, BMS College of Engineering, Bangalore during the academic semester Sep-Dec 2025. We also declare that to the best of our knowledge and belief, the project reported here is not from part of any other report by any other students.

Signature of the Candidates

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PROBLEM STATEMENT

In many small and medium-scale businesses, billing operations are still performed manually, which is time-consuming and prone to errors. Manual billing systems require significant effort in calculating totals, taxes, and generating invoices, leading to inaccuracies, data loss, and poor record management. Maintaining customer and transaction details on paper also makes data retrieval difficult and inefficient.

There is a need for an automated billing system that can efficiently manage product details, calculate bills accurately, generate invoices quickly, and store transaction records securely. The proposed Billing System aims to overcome these limitations by providing a computerized solution that improves accuracy, reduces manual effort, saves time, and ensures reliable data management using Object-Oriented Programming concepts.

INTRODUCTION

Billing is one of the most important operations in any business, as it directly affects revenue management, customer satisfaction, and record keeping. In many organizations, especially small and medium enterprises, billing is still handled manually or using basic systems that lack efficiency and accuracy. Manual billing processes involve repetitive calculations, handwritten invoices, and physical storage of records, which often leads to errors, delays, and difficulty in tracking past transactions.

With the rapid advancement of technology, there is a growing need for automated systems that can simplify business operations. An automated Billing System helps in generating bills quickly and accurately while maintaining proper records of products, customers, and transactions. Such a system reduces human intervention, minimizes calculation errors, and ensures consistency in billing operations.

This project aims to design and develop a Billing System using Object-Oriented Programming principles. The system is structured into modules that handle product management, billing calculation, invoice generation, and data storage. OOP concepts such as encapsulation, inheritance, and polymorphism are used to make the system modular, reusable, and easy to maintain. The proposed system provides a user-friendly interface, improves operational efficiency, and offers a reliable solution for modern billing requirements.

Overview of the project

User Interface

(Customer details, product selection, quantity input)



Processing Module

(Bill calculation, tax, discount, validation)



Database / Storage

(Product data, customer records, transaction details)



Bill / Invoice Output

(Display or print final bill)

TOOLS USED

The development of the Billing System requires various software tools to design, implement, and test the application efficiently. The following tools were used in this project:

- **Programming Language:** Java
Used to implement the core logic of the billing system using Object-Oriented Programming concepts.
- **Integrated Development Environment (IDE):** Eclipse / NetBeans
Used for writing, compiling, and debugging the source code.
- **Database:** MySQL
Used to store product details, customer information, and billing records securely.
- **Operating System:** Windows
Provides the platform for development and execution of the application.
- **Other Tools:** Command Prompt / Browser
Used for program execution and database access.

OOPS CONCEPT AND ITS EXPLANATION

Object-Oriented Programming (OOP) is a programming approach that organizes a system using objects and classes. It improves code reusability, security, and maintainability. The following OOPS concepts are used in the Billing System:

1. Class and Object

A **class** is a blueprint that defines properties and methods, while an **object** is an instance of a class. In this project, classes are created for products, customers, and bills, and objects represent their real-time data.

2. Encapsulation

Encapsulation is the process of binding data and methods into a single unit and restricting direct access to data. In the billing system, variables such as price and quantity are kept private and accessed through public methods to ensure data security.

3. Inheritance

Inheritance allows one class to acquire properties and methods of another class. The billing system uses inheritance to reuse common functionalities, reducing code duplication and improving efficiency.

4. Polymorphism

Polymorphism allows the same method to perform different tasks based on input or object type. It is used in the billing system to calculate bills differently for various products or customers.

5. Abstraction

Abstraction hides internal implementation details and shows only essential features. The billing system uses abstraction to expose only required operations to the user while hiding complex calculations.

IMPLEMENTATION

```
import java.util.Scanner;

import java.text.SimpleDateFormat;

import java.util.Date;

import java.util.Random;


class Item {

    String name;

    int quantity;

    double price;

    double gstRate;


    Item(String name, int quantity, double price, double gstRate) {

        this.name = name;

        this.quantity = quantity;

        this.price = price;

        this.gstRate = gstRate;

    }


    double getTotal() {

        return quantity * price;

    }


    double getGST() {

        return getTotal() * gstRate;

    }

}
```

```
public class BillingSystem {

    static Item[] items = new Item[50];

    static int count = 0;

    // GST CATEGORY SELECTION
    static double selectGSTCategory(Scanner sc) {

        System.out.println("\nSelect Item Type:");

        System.out.println("\n1. Essential Goods (0% GST)");
        System.out.println("    — Basic daily necessities");
        System.out.println("Examples:");
        System.out.println("- Fruits, vegetables");
        System.out.println("- Rice, wheat, milk, eggs");
        System.out.println("- Education and healthcare");
        System.out.println("- Public transport");

        System.out.println("\n2. Merit Goods (5% GST)");
        System.out.println("    — Daily-use household items");
        System.out.println("Examples:");
        System.out.println("- Tea, coffee");
        System.out.println("- Sugar, oil, spices");
        System.out.println("- Food grains, butter, ghee");
        System.out.println("- Clothes & footwear below ₹1000");
        System.out.println("- Domestic LPG, life-saving medicines");
        System.out.println("- Agarbatti, fertilizers");

        System.out.println("\n3. Standard Goods (18% GST)");
```

```
System.out.println("— Common consumer goods and services");
System.out.println("Examples:");
System.out.println("- Mobile phones, laptops, computers");
System.out.println("- TV, refrigerator, washing machine, AC");
System.out.println("- Internet, banking, insurance");
System.out.println("- Professional services");
System.out.println("- Branded clothes above ₹1000");
```

```
System.out.println("\n4. Luxury / Sin Goods (40% GST)");
System.out.println("— Luxury or harmful goods taxed heavily");
System.out.println("Examples:");
System.out.println("- Luxury cars, SUVs, high-end bikes");
System.out.println("- Cigarettes, tobacco, pan masala");
System.out.println("- Soft drinks");
System.out.println("- Online gaming and casinos");
```

```
System.out.print("\nEnter category number: ");
int choice = sc.nextInt();
```

```
switch (choice) {
    case 1: return 0.00;
    case 2: return 0.05;
    case 3: return 0.18;
    case 4: return 0.40;
    default:
        System.out.println("Invalid choice! Default GST = 18%");
        return 0.18;
}
}
```

```
static void addItem(Scanner sc) {  
  
    System.out.print("\nEnter item name: ");  
    String name = sc.next();  
  
    System.out.print("Enter quantity: ");  
    int qty = sc.nextInt();  
  
    System.out.print("Enter price per item: ");  
    double price = sc.nextDouble();  
  
    double gstRate = selectGSTCategory(sc);  
  
    items[count] = new Item(name, qty, price, gstRate);  
    count++;  
  
    System.out.println("\n ■ Item added successfully!\n");  
}
```

```
static void displayBill() {  
  
    if (count == 0) {  
        System.out.println("\n⚠ No items added!\n");  
        return;  
    }  
  
    double subTotal = 0, totalGST = 0;
```

```
String date = new SimpleDateFormat("dd-MM-yyyy").format(new Date());
```

```
int billNo = new Random().nextInt(9000) + 1000;
```

```
System.out.println("\n===== BILL =====");
```

```
System.out.println("Bill No : " + billNo);
```

```
System.out.println("Date   : " + date);
```

```
System.out.println("-----");
```

```
System.out.println("Item\tQty\tPrice\tGST%\tAmount");
```

```
for (int i = 0; i < count; i++) {
```

```
    Item it = items[i];
```

```
    double itemTotal = it.getTotal();
```

```
    double gst = it.getGST();
```

```
    subTotal += itemTotal;
```

```
    totalGST += gst;
```

```
System.out.printf("%s\t%d\t%.2f\t%.0f%%\t%.2f\n",
```

```
    it.name,
```

```
    it.quantity,
```

```
    it.price,
```

```
    it.gstRate * 100,
```

```
    itemTotal + gst
```

```
);
```

```
}
```

```
double discount = (subTotal > 10000) ? subTotal * 0.10 : 0;
```

```
double grandTotal = subTotal + totalGST - discount;

System.out.println("-----");
System.out.printf("Sub Total : Rs. %.2f\n", subTotal);
System.out.printf("Total GST : Rs. %.2f\n", totalGST);
System.out.printf("Discount : Rs. %.2f\n", discount);
System.out.println("-----");
System.out.printf("Grand Total: Rs. %.2f\n", grandTotal);
System.out.println("=====\\n");
}
```

```
public static void main(String[] args) {

    Scanner sc = new Scanner(System.in);
    int choice;

    do {
        System.out.println("===== BILLING SYSTEM =====");
        System.out.println("1. Add Item");
        System.out.println("2. Display Bill");
        System.out.println("3. Exit");
        System.out.print("Enter choice: ");
        choice = sc.nextInt();

        switch (choice) {
            case 1:
                addItem(sc);
                break;
            case 2:
```

```
        displayBill();

        break;
    case 3:

        System.out.println("👋 Thank you! Visit Again");

        break;
    default:

        System.out.println("⚠ Invalid choice!");
    }
} while (choice != 3);

sc.close();
}
}
```

RESULT

```
D:\javaprogram>java BillingSystem
===== BILLING SYSTEM =====
1. Add Item
2. Display Bill
3. Exit
Enter choice: 1

Enter item name: rice
Enter quantity: 5
Enter price per item: 60

Select Item Type:

1. Essential Goods (0% GST)
? Basic daily necessities
Examples:
- Fruits, vegetables
- Rice, wheat, milk, eggs
- Education and healthcare
- Public transport

2. Merit Goods (5% GST)
? Daily-use household items
Examples:
- Tea, coffee
- Sugar, oil, spices
- Food grains, butter, ghee
- Clothes & footwear below ₹1000
- Domestic LPG, life-saving medicines
- Agarbatti, fertilizers

3. Standard Goods (18% GST)
? Common consumer goods and services
Examples:
- Mobile phones, laptops, computers
- TV, refrigerator, washing machine, AC
- Internet, banking, insurance
- Professional services
- Branded clothes above ₹1000

4. Luxury / Sin Goods (40% GST)
? Luxury or harmful goods taxed heavily
Examples:
- Luxury cars, SUVs, high-end bikes
```

Enter category number: 1

? Item added successfully!

===== BILLING SYSTEM =====

1. Add Item
2. Display Bill
3. Exit

Enter choice: 1

Enter item name: tea

Enter quantity: 2

Enter price per item: 150

Select Item Type:

1. Essential Goods (0% GST)

? Basic daily necessities

Examples:

- Fruits, vegetables
- Rice, wheat, milk, eggs
- Education and healthcare
- Public transport

2. Merit Goods (5% GST)

? Daily-use household items

Examples:

- Tea, coffee
- Sugar, oil, spices
- Food grains, butter, ghee
- Clothes & footwear below ₹1000
- Domestic LPG, life-saving medicines
- Agarbatti, fertilizers

3. Standard Goods (18% GST)

? Common consumer goods and services

Examples:

- Mobile phones, laptops, computers
- TV, refrigerator, washing machine, AC
- Internet, banking, insurance
- Professional services
- Branded clothes above ₹1000

4. Luxury / Sin Goods (40% GST)

Enter category number: 2

? Item added successfully!

===== BILLING SYSTEM =====

1. Add Item
2. Display Bill
3. Exit

Enter choice: 1

Enter item name: mobile

Enter quantity: 2

Enter price per item: 15000

Select Item Type:

1. Essential Goods (0% GST)

? Basic daily necessities

Examples:

- Fruits, vegetables
- Rice, wheat, milk, eggs
- Education and healthcare
- Public transport

2. Merit Goods (5% GST)

? Daily-use household items

Examples:

- Tea, coffee
- Sugar, oil, spices
- Food grains, butter, ghee
- Clothes & footwear below ₹1000
- Domestic LPG, life-saving medicines
- Agarbatti, fertilizers

3. Standard Goods (18% GST)

? Common consumer goods and services

Examples:

- Mobile phones, laptops, computers
- TV, refrigerator, washing machine, AC
- Internet, banking, insurance
- Professional services
- Branded clothes above ₹1000

4. Luxury / Sin Goods (40% GST)
? Luxury or harmful goods taxed heavily
Examples:

- Luxury cars, SUVs, high-end bikes
- Cigarettes, tobacco, pan masala
- Soft drinks
- Online gaming and casinos

Enter category number: 3

? Item added successfully!

===== BILLING SYSTEM =====

1. Add Item
2. Display Bill
3. Exit

Enter choice: 1

Enter item name: cigarette

Enter quantity: 3

Enter price per item: 300

Select Item Type:

1. Essential Goods (0% GST)

? Basic daily necessities

Examples:

- Fruits, vegetables
- Rice, wheat, milk, eggs
- Education and healthcare
- Public transport

2. Merit Goods (5% GST)

? Daily-use household items

Examples:

- Tea, coffee
- Sugar, oil, spices
- Food grains, butter, ghee
- Clothes & footwear below ₹1000
- Domestic LPG, life-saving medicines
- Agarbatti, fertilizers

3. Standard Goods (18% GST)

4. Luxury / Sin Goods (40% GST)
? Luxury or harmful goods taxed heavily

Examples:

- Luxury cars, SUVs, high-end bikes
- Cigarettes, tobacco, pan masala
- Soft drinks
- Online gaming and casinos

Enter category number: 4

? Item added successfully!

===== BILLING SYSTEM =====

1. Add Item
2. Display Bill
3. Exit

Enter choice: 2

===== BILL =====

Bill No : 1493

Date : 05-01-2026

Item	Qty	Price	GST%	Amount
rice	5	60.00	0%	300.00
tea	2	150.00	5%	315.00
mobile	2	15000.00	18%	35400.00
cigarette	3	300.00	40%	1260.00

Sub Total : Rs. 31500.00

Total GST : Rs. 5775.00

Discount : Rs. 3150.00

Grand Total: Rs. 34125.00

===== BILLING SYSTEM =====

1. Add Item
2. Display Bill
3. Exit

Enter choice: 3

? Thank you! Visit Again

D:\javaprogram>java BillingSystem

===== BILLING SYSTEM =====

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- Herbert Schildt, *Java: The Complete Reference*, McGraw Hill.
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- Lecture Notes and Study Materials provided by the Institution.
- Online tutorials and programming resources for Java and OOP concepts.